N-C-521E
March 3, 1970
SUPERSEDING
Int. Fed. Spec. N-C-00521D(AGR-C&MS)
July 5, 1966 and
Fed. Spec. N-C-521C
May 19, 1960

FEDERAL SPECIFICATION

CORN MEAL (REGULAR AND DEGERMED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for corn meal for use by agencies of the Federal Government.

1.2 Classification.

1.2.1 Types, classes, granulations, and colors. Corn meal covered by this specification shall be of the following types, classes, granulations, and colors, as specified (see 6.1):

Type I - Regular corn meal.
Type II - Degermed corn meal.

Class A - Plain.
Class B - Enriched.
Granulation 1 - Coarse.
Granulation 2 - Fine.
Color a - White.
Color b - Yellow.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-S-48 - Sacks, Shipping, Paper.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-C-29 - Canned Subsistence Items, Packaging and Packing of.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8920

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Military Specifications:

MIL-C-1470 - Cereals, Breakfast, Prepared (Ready-to-eat) and Uncooked; Packaging, Packing, Labeling, and Marking of.
MIL-T-21330 - Treatment, Insect-Resistant, for Paper.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Establishments.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Governmental:

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Prices may be obtained from the Superintendent of Documents.)

Official Grain Standards of the United States, SRA-C&MS 177.
United States Standards for Condition of Food Containers.

(Copies may be obtained from the Information Division, Consumer and Marketing Service, U. S. Department of Agriculture, Washington, D. C. 20250.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Copies may be obtained from the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

American Association of Cereal Chemists Approved Methods.

(Copies may be obtained from the American Association of Cereal Chemists, 1821 University Avenue, St. Paul, Minnesota 55104.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Attn: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Material. White corn meal and yellow corn meal shall be prepared from white corn or yellow corn, as applicable, both as defined in the Official Grain Standards of the United States for Corn. The corn shall contain not more than 4.0 percent of broken corn and foreign material and not more than 5.0 percent of damaged kernels, of which not more than 0.2 percent shall be heat-damaged kernels. Corn which is musty, sour, or heating; or which has any commercially objectionable foreign odor; or which is otherwise of distinctly low quality shall not be used in the production of regular or degermed corn meal.

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3.1.1 Type I. Regular corn meal shall be produced by grinding corn into granules in accordance with 3.4.

3.1.2 Type II. Degermed corn meal shall be produced by grinding degermed, hulled corn into granules in accordance with 3.4.

3.2 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.3 Chemical requirements. Unless otherwise specified, regular corn meal and degermed corn meal shall conform to the chemical requirements shown in table I.

TABLE I. Chemical requirements

Type	Minimum limits of	Maximum limits of		
	Fat 1/	Ash 1/	Fat 1/	Moisture
	Percent	Percent	Percent	Percent
I	3.5	1.80	---	13.5
II	---	.60	1.5	13.5

1/These limiting values are on a moisture-free basis.

3.4 Granulation.

3.4.1 Granulation 1 (type I or II). Unless otherwise specified, coarse regular corn meal and coarse degermed corn meal shall consist of particulate granules of such size that not less than 99 percent will pass through a U.S. No. 20 woven-wire-cloth sieve, not less than 75 percent will pass through a U.S. No. 25 woven-wire-cloth sieve, not more than 30 percent will pass through a U. S. No. 45 woven-wire-cloth sieve, and not more than 10 percent will pass through a U. S. No. 80 woven-wire-cloth sieve (see 4.3.1.2).

3.4.2 Granulation 2 (type I or II). Fine regular corn meal and fine degermed corn meal shall consist of particulate granules of such size that not less than 99 percent will pass through a No. 20 woven-wire-cloth sieve, not less than 90 percent will pass through a U. S. No. 25 woven-wire-cloth sieve, not less than 30 percent will pass through a U. S. No. 45 woven-wire-cloth sieve, and not more than 20 percent will pass through a U. S. No. 80 woven-wire-cloth sieve (see 4.3.1.2).

3.5 Color, taste, and odor requirements. The product shall have a good characteristic color; possess a natural ground corn taste and odor; and be free from rancid, bitter, musty, and other undesirable tastes and odors.

3.6 Enrichment.

3.6.1 Type I, class B, and Type II, class B. Enriched regular corn meal and degermed corn meal shall contain thiamine (vitamin B_1), riboflavin (vitamin B_2), niacin or niacinamide, and iron (in harmless and assimilable form) in the following proportions:

Enrichment ingredient	Minimum mg. per lb.	Maximum mg. per lb.
Thiamine	2.0	3.0
Riboflavin	1.2	1.8
Niacin or niacinamide	16.0	24.0
Iron	13.0	26.0

3.6.1.1 In addition to the required enrichment ingredients under 3.6.1, enriched regular corn meal and enriched degermed corn meal may contain vitamin A, vitamin D, and calcium (in harmless and assimilable form) as optional ingredients in the following proportions:

	Minimum	Maximum
Vitamin A Palmitate - I. U. per lb. 1/	4,000	6,000
Vitamin D - U. S. P. Units per lb.	250	1,000
Calcium - mg. per lb.	500	750

1/Vitamin A Palmitate shall be added in encapsulated or beadlet form. Particle size of the beadlet shall comply with the requirement that 100 percent shall pass through a U. S. Standard No. 30 sieve. The product shall be no less than 95 percent of the all-transisomer as determined

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by the U.S.P. assay procedure. Corn meal should not be sifted after vitamin supplementation through a cloth which would remove the vitamin A beadlets.

The use of the vitamin A Palmitate preparation shall contribute no off flavor or odor to the dry mix, or to the cooked products prepared from the fortified corn meal.

3.7 Workmanship. The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. For military agencies the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. At least one sample for analysis shall be taken from each lot. Each sample shall consist of at least 1 quart and shall be representative of the entire lot. The sample shall be placed in a moisture-proof container at the time the sample is taken. The container shall be completely filled with the sample and the sample shall be retained in the container until it is to be tested. Samples of finished products covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished products may be taken at any time by the Government inspector to determine compliance with the requirements.

4.2 Inspection.

4.2.1 General. Unless otherwise specified, inspection shall be made by a representative of the purchasing agency or the United States Department of Agriculture at the time and place designated by the purchasing agency. It may be made at the site of preparation, both during and after processing, at a suitable point in transit, or after delivery to the point of destination. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.2 Condition inspection of containers. When inspection of product for quality and condition is performed by the U. S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for military agencies, see 4.2.3.5.)

4.2.3 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examinations and tests prescribed herein, using his own or other inspection facilities and services acceptable to the Government.

4.2.3.1 Availability of records. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.3.2 Preacceptance inspection (plant sanitation inspection).

4.2.3.2.1 For civil agencies. Inspection of plants producing the product will be conducted upon the request of the purchasing and contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.3.2.2 For military agencies. Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.2.3.3 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.3.4 Component and material inspection. In accordance with 4.2.3 above, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.3.4.1 Corn. Conformance of corn to identity and condition requirements of 3.1 shall be ascertained by review of pertinent certificate(s) issued by an official inspection agency. Use of nonconforming components shall be cause for rejection of the involved end item. When corn that has not been officially inspected is used, the miller will certify that he has inspected it against standards equivalent to those used by a licensed inspection agency and has found that it does conform.

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4.2.3.5 Examination of end item. Classification of defects found during examination shall be in accordance with tables III through V and 4.2.3.5.1 through 4.2.3.5.5. Sampling plans and acceptance criteria for the aforementioned tables shall be as shown in table II. Acceptable quality levels (AQLs) shall be expressed as percent defectives for table III and in terms of defects per hundred units for tables IV and V.

TABLE II. End item examination criteria

Table	Inspection level		Lot size expressed in	AQLs		
				Major	Minor	Total
III	S-3	1 filled primary container	Primary container	---	2.5	----
IV	I	1 filled and closed carton	Cartons	1.5	---	4.0
V	S-2	Contents of 1 primary container	Primary container	1.5	---	----

TABLE III. Examination of net weight 1/ 2/

Category	Defect
Minor	
201	More than 2 percent under required net weight.

1/Report results to the nearest 1/4 ounce.
2/Lot shall be rejected if sample data indicate a lot average net weight less than required net weight.

TABLE IV. Examination of cartons

Category		Defect
Major	Minor	
101		Tear or hole through carton.
102		Sifter.
103		Nomenclature missing, incorrect, or illegible.
104		Not clean.
	201	Other required label information missing, incorrect, or illegible.

TABLE V. Examination of product characteristics 1/

Category	Defect
Major	
101	Lumpy or caked (cannot be broken by light hand pressure).
	Color
102	Not characteristic (yellow or white, as applicable). 2/
103	Off color (grayish).
104	Naturally discolored by soot or black specks due to heat damage.
	Flavor and odor
105	Does not possess a natural ground corn flavor and odor.
106	Rancid, bitter, sour, or otherwise objectionable.

1/Presence of foreign material (e.g., filth, insect(s) or insect fragments, glass, metal, stones, paint, wood, etc.) shall be cause for rejection of the lot. (Examine visually by using Hastings 3X lens or equivalent.)
2/A slight tinge of light straw color or of pink shall not affect classification as white. A slight tinge of red shall not affect classification as yellow.

4.2.3.5.1 Examination of cans. Examination of cans shall be in accordance with the applicable quality assurance provisions of MIL-C-1470.

4.2.3.5.2 Examination of can labeling. Examination shall be in accordance with the applicable quality assurance provisions of MIL-C-1470.

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4.2.3.5.3 Examination of packing, marking, waterproof case liners, and shipping containers for cartons. Examination shall be in accordance with the applicable quality assurance provisions of MIL-C-1470. Shipping containers with marked net weight exceeding 48 pounds shall not be acceptable.

4.2.3.5.4 Examination for packing and shipping container for cans. Examination shall be in accordance with the applicable quality assurance provisions of PPP-C-29.

4.2.3.5.5 Examination of shipping sacks. Filled and closed shipping sacks shall be examined in accordance with the quality assurance provisions of the end item as contained in UU-S-48. In addition, the following defect shall be included in the examination: Major: Markings missing, incorrect, or illegible. Sacks required to be insect-resistant treated shall not be acceptable in the event the date of fabrication of the sacks indicates that the sacks will be more than 60 days old at the time of shipment of the corn meal.

4.2.3.5.6 Examination for closure of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure shall be in accordance with the appendix of that specification.

4.2.4 Sampling procedures and acceptance criteria for testing of finished product. The finished product shall be tested for the chemical characteristics in table I, for filth (see 3.2), for granule size as specified in 3.4.1 and 3.4.2, and for enrichment components specified in 3.6, as applicable. Test procedures shall be in accordance with 4.3. The filth test shall be performed by the Government. The sample for testing shall be a 2-pound composite derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of the primary container. One determination shall be made for granule size. Results shall be reported to the same decimal place or unit as specified for the pertinent requirements, except that vitamin D, when applicable, shall be reported to the nearest 50 U.S.P. units and the report for filth on a pass or fail basis. The lot of product shall be rejected if test results indicate nonconformance to one or more of the aforementioned characteristics.

4.2.5 Standby test samples. The Government reserves the right to withdraw and hold test samples of components or finished product or both (quantity of which shall be not more than twice that specified by this specification) for inspection purposes. These samples, if not used, will be returned to the vendor.

4.2.6 Final inspection. Unless otherwise specified, final inspection shall be made after delivery to the point of destination (see 6.2).

4.3 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.3.1 Examination procedures. Unless otherwise specified herein, examinations specified in 4.3.1.1 and 4.3.1.2 shall be made on the basis of a representative portion of approximately 100 grams taken from the original sample.

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4.3.1.1 Organoleptic examinations. The color, taste, and odor shall be determined by visual inspection, testing, smelling, or general examination of the sample.

4.3.1.2 Sieving test. U. S. Standard No. 20, No. 25, No. 45, and No. 80 woven-wire-cloth sieves which are 8 inches in diameter and which have full height frames, shall be used in making the sieving test. A cover and bottom pan to fit the sieves are also required. Place a representative 20.0 gram sample of the corn meal on the No. 20 sieve, attach the cover and bottom pan, hold the sieve assembly in a slightly inclined position with one hand and shake by striking the side of the sieve assembly against the other hand with a short upward stroke at the rate of about 150 times per minute, turning the sieve assembly about 1/6 of a revolution, each time in the same direction, after each 25 strokes. Continue shaking for 2 minutes. Remove material from the bottom pan and weigh it. Attach the bottom pan to the No. 25 sieve. Place the material that passed through the No. 20 sieve on the No. 25 sieve and repeat the above sieving procedure. Attach the bottom pan to the No. 45 sieve and place the material that passed through the No. 25 sieve on the No. 45 sieve. With the assembly on a level surface and using a flat 1-inch varnish brush, bring the brush lightly over the diameter of the sieve at the rate of 140 strokes per minute, rotating the assembly after each few strokes. Continue brushing for 1 minute always working from the far side of the sieve toward the operator. Remove the material from the bottom pan and weigh it. Using the material that passed through the No. 45 sieve, repeat the same brushing operation on the No. 80 sieve. From the weights determined, calculate the percentages of the original sample that passed through each of the four sieves.

4.3.2 Test procedures.

4.3.2.1 Chemical analyses. Chemical analyses required to determine the fat, ash, moisture content, and enrichment shall be made in accordance with the applicable methods prescribed in Official Methods of Analysis of the Association of Official Agricultural Chemists or American Association of Cereal Chemists Approved Methods in effect on the date of invitation for bids.

4.3.2.2 Filth tests. All tests required to determine compliance with the provisions of the Federal Food, Drug, and Cosmetic Act, in regard to rodent excreta, rodent hairs, insects, and insect fragments, shall be made in accordance with the methods referred to in 4.3.2.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Military agencies. The product shall be packaged in accordance with MIL-C-1470 for the levels specified (see 6.1).

5.1.2 Civil agencies. Packaging shall be level B or C as specified (see 6.1).

5.1.2.1 Level B, twenty-ounce units. Twenty ounces of corn meal shall be packaged in a box conforming to PPP-B-566, style 1, type A, class b. When specified (see 6.1), the box shall be lined with a glassine, kraft, or kraft-glassine laminated bag or tube with open ends closed to obtain a full sift-proof inner package.

5.1.2.2 Level C. Corn meal shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 Military agencies. The product shall be packed in accordance with MIL-C-1470 for the levels specified (see 6.1).

5.2.2 Civil agencies. Packing shall be level B or C as specified (see 6.1).

5.2.2.1 Level B, twenty-ounce units. Twenty-four-unit packages of corn meal packaged as specified in 5.1. shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.2.1.1 Twenty-four- and one hundred-pound units. Twenty-four or one hundred pounds of corn meal of the type and class specified (see 6.1), shall be packed into a multiwall shipping sack fabricated and closed in accordance with type I, class 2, style B, or type II, style A, construction 4-4X or 9-9X, respectively, of UU-S-48. The outer surface of the sack shall be insect-resistant treated in accordance with MIL-T-21330.

5.2.2.2 Level C. Corn meal, in quantities as specified (see 6.1), and packaged as specified (see 5.1), shall be packed in containers to assure carrier acceptance and safe arrival at destination

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in complicance with National Motor Freight Classification Rules or Uniform Freight Classification Rules, as applicable.

5.3 Labeling and marking.

5.3.1 Civil agencies.

5.3.1.1 Unit containers. Any commercial labeling and additional labeling as specified, that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.3.1.2 Shipping containers. Shipping containers shall be marked in accordance with Fed. Std. No. 123, and UU-S-48, as applicable, and such other pertinent information as specified by individual agencies.

5.3.2 Military agencies.

5.3.2.1 Cans. The cans shall be plainly lithograped on the body or end in black letters in accordance with MIL-C-1470 as specified, and as follows:

CORN MEAL, REGULAR, TYPE I	or	CORN MEAL, DEGERMED, TYPE II
(In letters larger than any others used for the label)		Class (Plain or Enriched) Granulation (Coarse or Fine) Color (Yellow or White) ----------LBS. NET

5.3.2.2 Twenty-four-ounce package. Unless otherwise specified, commercial labeling shall be acceptable.

5.3.2.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, granulation, and color of regular corn meal or degermed corn meal required (see 1.2.1).
(c) Selection of the applicable levels of packaging and packing (see 5.1).
(d) Quantity required and net weight of containers (see 5.1).
(e) Optional enrichment when required (see 3.6.1.1).

6.2 Destination inspection (for civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only. Unless otherwise specified, such inspection should be made by the receiving facility.

6.3 If samples are required with bid, this requirement should be stated, giving information as to size or quantity of sample, and the particular purpose to be served by the bid sample.

6.4 Federal specifications do not include all types, classes, grades, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - GL, MD
Navy - SA, MC
Air Force - 45

Preparing activity:

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CIVIL AGENCIES INTEREST:

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